Karl Kaspar Haavel

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Estonia | Tallinn | Estonian/English

Data Scientist with a track record of developing and deploying forecasting and computer vision models in production. Looking for roles to push boundaries and create real impact.

Experience

5/2024-12/2025 **Renewcast SRL** Senior Data Scientist
Time-series analysis. Built and deployed renewables power forecasting models actively used in balance market trading across multiple power markets and clients using diverse data sources.

9/2021-4/2024 **Proekspert AS** Data Scientist
Enlighten (ARIANE) - Time series analysis. Fault detection and localisation for a reusable rocket engine. Progressed the project from TRL2 to TRL5, demonstrating readiness for operational use.
THOR - Computer Vision. Satellite data-based roof detection (shape/size/pitch) to speed up sales processes. Client reported accelerated sales and a significant increase in inventory turnover.
City of Oslo - Computer Vision. Was apart of building and deploying a methodologies for detecting, tracking and localising traffic signs for automated pre-labelling.

7/2018-8/2020 **Socar Trading SA** Middle Officer
Risk management. Development and maintenance of market risk and valuation models for oil&gas trading. Exposure and P&L reporting for various trading desks across the world.

3/2018-6/2018 **Socar Trading Services AS** Junior Middle Officer

6/2016-12/2016 **Eesti Energia AS**. Financial Analyst
Internship: Fundamental analysis of multiple energy markets, creating statistical models to help traders make investment decisions.

Education

9/2020-6/2022 **Tartu University** Cum Laude. Faculty of Computer Science. MsC. Data Science.
Thesis: "Exploring Out-of-Distribution Detection using Vision Transformers"
Publication: "Probability Density from Latent Diffusion Models for Out-of-Distribution Detection"

9/2013-6/2016 **Tartu University** Faculty of Economics. BSS. Economics.
Thesis: "Expiration Effect of Currency Options on the Currency Market: The Example of EUR/CAD"

9/2001-6/2013 **Gustav Adolf Grammar School**

Software

Expert	Python (Pytorch, Numpy, Scipy, Pandas, Xarray, OpenCV, MLflow), SQL, Docker
Advanced	Azure, Spark, DataBricks, GIT, Kubeflow
Beginner	C++ (CUDA), Kubernetes, Google Cloud

Projects (Public)

Pattern Based Trading with Similarity Measures	Repository | Poster
Transformers and CNN in analysing microscopy images	Repository | Blog
Audio Thumbnail	Repository | Website
Extractive Text Summarisation Methods in Estonian	Report | Poster